

JEE MAIN REVISION MASTER: HEAT AND THERMODYNAMICS 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Carnot Engine & Second Law Efficiency	12 PYQs	Heat Engines
High	First Law & Thermodynamic Processes	10 PYQs	First Law
Moderate	Kinetic Theory of Gases & Thermal Radiation	7 PYQs	KTG & Radiation

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Carnot Engine & Second Law Efficiency	$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$, $\text{COP} = \frac{1}{\eta}$ $\frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$, $\eta = \frac{1 - \eta}{\eta}$
First Law & Thermodynamic Processes	$\Delta Q = \Delta U + W$, $\Delta U = n C_v \Delta T$, $W_{\text{iso}} = nRT \ln \left(\frac{V_2}{V_1} \right)$, $W_{\text{adi}} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$
Kinetic Theory of Gases & Thermal Radiation	$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$, $v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$, $\lambda_m T = b$, $E = e \sigma A T^4$, $\frac{T_1 - T_2}{t} = K \left(\frac{T_1 + T_2}{2} - T_0 \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Carnot Engine & Second Law Efficiency (High Yield — 12 PYQs)

Core Principle: Carnot cycle efficiency limit, reversible refrigerators, and Clausius/Kelvin-Planck statements.

■ **Caution / JEE Pitfall:** Temperatures T_1 (source) and T_2 (sink) MUST strictly be substituted in Kelvin (K), never in Celsius ($^{\circ}\text{C}$).

■ **Speed Trick:** To maximize Carnot efficiency, lowering sink temperature T_2 by ΔT is more effective than raising source temperature T_1 by the same ΔT .

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Three metal rods of the same material and identical in all respects are joined as shown in the figure. The temperatures at the ends of these..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "One mole of an ideal monoatomic gas is taken round the cyclic process MNOM. The work done by the gas is..." [Ans: N/A]

3.2 First Law & Thermodynamic Processes (High Yield — 10 PYQs)

Core Principle: First law of thermodynamics, molar heat capacities ($C_p - C_v = R$), and work done in cyclic/open processes.

■ **Caution / JEE Pitfall:** Internal energy $\Delta U = nC_v\Delta T$ depends ONLY on temperature change ΔT , regardless of whether the path is isobaric, isothermal, or adiabatic.

■ **Speed Trick:** In cyclic processes on a P-V diagram: clockwise cycles give positive net work (heat engine); counter-clockwise cycles give negative work (refrigerator).

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "In thermodynamic processes, which of the following statements is not true?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "The graph of pressure P and volume V of 1 mole of an ideal gas at constant temperature is..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "A gas is taken from state A to state B along two different paths 1 and 2. The heat absorbed and work done by the system along these two path..." [Ans: N/A]

3.3 Kinetic Theory of Gases & Thermal Radiation (Moderate Yield — 7 PYQs)

Core Principle: Molecular speeds, energy equipartition $E = (f/2)nRT$, Stefan-Boltzmann law, Wien's displacement law, and Newton's law of cooling.

■ **Caution / JEE Pitfall:** Degrees of freedom: Monatomic $f = 3$ ($\gamma = 5/3$); Diatomic at room temp $f = 5$ ($\gamma = 7/5$); Polyatomic non-linear $f = 6$ ($\gamma = 4/3$).

■ **Speed Trick:** Net heat radiated by a body at temperature T in surroundings T_0 is $P_{\text{(net)}} = e\sigma A(T^4 - T_0^4)$.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "At 27°C temperature, the mean kinetic energy of the atoms of an ideal gas is E_1 . If the temperature is increased to 327°C ..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "100g of ice at 0°C is mixed with 100g of water at 100°C . The final temperature of the mixture is $[Take, L_f = 3.36 \text{ kJ kg}^{-1}]$..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Heat cannot flow itself from a body at lower temperature to a body at higher temperature. This statement corresponds to..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and air ignited by a spark III. The leaking of gas from the kitchen cylinder. IV. The transfer of heat from one heated part of a liquid to the other colder part. Which amongst these processes are irreversible in nature?

- (A) I and II
- (B) I and III
- (C) II, III and IV
- (D) I, II, III and IV

Correct Answer: None

Solution Analysis: All the given processes are irreversible in nature. Free expansion of a gas is irreversible because the gas spreads into the available space on its own. It cannot return to its original volume without some external work being done. Combustion of petrol and air is also irreversible. Once petrol burns, it forms new substances like carbon dioxide, water vapour, etc. These products cannot change back into petrol and air by simply reversing the process. Leaking of gas from a kitchen cylinder is irreversible because the gas moves from high pressure inside the cylinder to lower pressure outside. It w

Benchmark Problem 2: KCET 2026 (Q#2)

In thermodynamic processes, which of the following statements is not true?

- (A) In an isothermal process, the temperature remains constant
- (B) In an isobaric process, the volume remains constant
- (C) In an adiabatic process, the system is insulated from the surroundings
- (D) In an adiabatic process, $PV^\gamma = \text{a constant}$

Correct Answer: None

Solution Analysis: The statement that is not true is: Option B In an isobaric process, the pressure remains constant, not the volume. The word "isobaric" means constant pressure. So for an isobaric process, $P = \text{constant}$ If the volume remains constant, that process is called an isochoric process. The other statements are correct: In an isothermal process, $T = \text{constant}$ In an adiabatic process, no heat is exchanged with the surroundings, so the system is insulated: $Q = 0$ Also, for a reversible adiabatic process of an ideal gas, $PV^\gamma = \text{constant}$ So the incorrect statement is Optio

Benchmark Problem 3: KCET 2025 (Q#2)

At 27°C temperature, the mean kinetic energy of the atoms of an ideal gas is E_1 . If the temperature is increased to 327°C, then the mean kinetic energy of the atoms will be

- (A) $\frac{E_1}{\sqrt{2}}$
- (B) $\sqrt{2}E_1$
- (C) $2E_1$
- (D) $\frac{E_1}{2}$

Correct Answer: None

Solution Analysis: The mean kinetic energy of the atoms in an ideal gas is directly proportional to the temperature measured in Kelvin. At an initial temperature of 27°C, the mean kinetic energy is E_1 . When the temperature is increased to 327°C, we can determine the new mean kinetic energy E_2 using the relationship:

$\frac{E_1}{E_2} = \frac{T_1}{T_2}$ To find E_2 , we can rearrange this equation: $E_2 = E_1 \frac{T_2}{T_1}$ Converting the given temperatures from Celsius to Kelvin: Initial temperature, $T_1 = 27^\circ\text{C} + 273 = 300\text{K}$ Final temperature, $T_2 = 327^\circ\text{C} + 273 = 600\text{K}$ Substitu